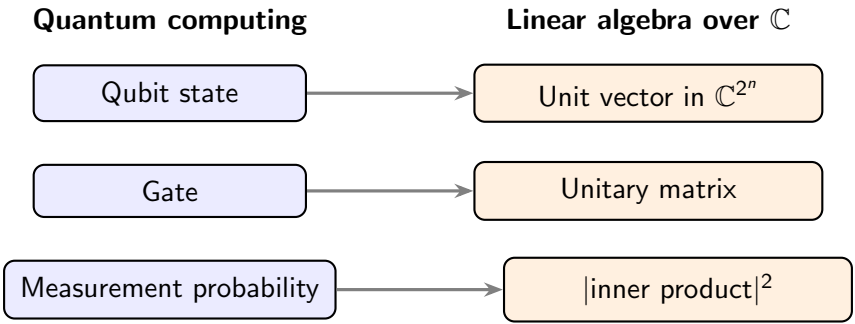
Linear Algebra Refresher for Quantum Computing

Quantum Computing and AI

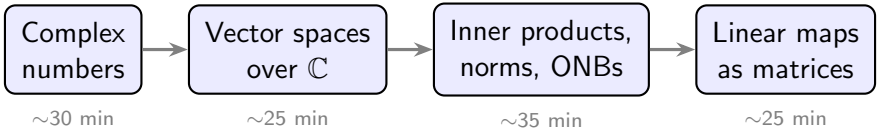
Lecturer: Amit Kumar

Department of Computer Science and Engineering
IIT Delhi

Why a Linear-Algebra Lecture?



Today's Plan



- Pictures are drawn in $\mathbb{R}^2$ for intuition; the algebra works in any $\mathbb{C}^n$.

Notation We Will Use

Object	Notation	Example
Scalar (complex number)	plain italic	$z,\ \alpha,\ \beta$
Complex conjugate	$\overline{z}$ (physics books: z^*)	$\overline{3+2i} = 3-2i$
Vector (column)	**bold**	$\mathbf{u},\ \mathbf{v},\ \psi$
Standard basis	$\mathbf{e}_1, \ldots, \mathbf{e}_n$	$\mathbf{e}_1 = (1,0)^T$
Conjugate transpose	$\mathbf{v}^\dagger,\ A^\dagger$	"v dagger"
Inner product	$\langle \mathbf{u}, \mathbf{v} \rangle$	a complex number
Norm (length)	$\|\mathbf{v}\|$	a non-negative real

Why Complex Numbers?

- $x^2 + 1 = 0$ has no real solution.

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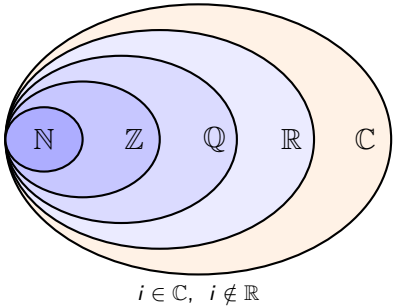
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- Invent a number i with $i^2 = -1$ and the problem disappears.
- In QC, amplitudes are complex: they carry a **magnitude** *and* a **phase**.
- Phases let amplitudes *cancel*: this is interference, the engine of quantum algorithms.



Complex Numbers: Definition

Definition

A complex number is $z = a + b\,i$ with $a, b \in \mathbb{R}$ and $i^2 = -1$.

$$\operatorname{Re} z = a, \qquad \operatorname{Im} z = b.$$

Two complex numbers are equal iff their real parts and imaginary parts are equal.

Examples

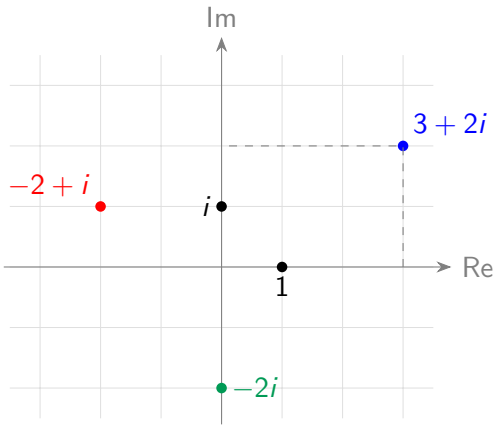
$z = 3 + 2i$: $\operatorname{Re} z = 3$, $\operatorname{Im} z = 2$. $w = -i$: $\operatorname{Re} w = 0$, $\operatorname{Im} w = -1$. $5 = 5 + 0i$ is also complex.

Note: $\operatorname{Im} z$ *is the real number* b*, not* bi*. A common slip.*

The Complex Plane

- Identify $z = a + bi$ with the point (a, b).
- Horizontal axis: real part.
  Vertical axis: imaginary part.
- So $\mathbb{C}$ "looks like" $\mathbb{R}^2$, but it also has a *multiplication*.

Also called the Argand diagram.



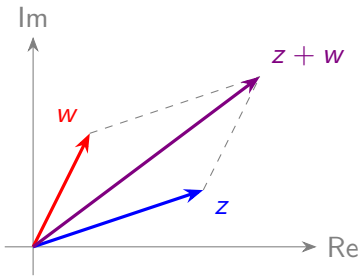
Addition = Adding Arrows

$$(a + bi) + (c + di) = (a + c) + (b + d)\, i$$

Example

$(3 + 2i) + (1 - 4i) = 4 - 2i$

Geometrically: the parallelogram rule, exactly like vectors in $\mathbb{R}^2$.



Multiplication

Expand as usual, then use $i^2 = -1$:

$$(a + bi)(c + di) = ac + adi + bci + bd\, i^2 = (ac - bd) + (ad + bc)\, i.$$

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Powers of i cycle with period 4

$$i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1, \quad i^5 = i, \ \ldots$$

Keep the cycle in mind: we will see it again as rotation by 90°.

Conjugate and Modulus

Definitions

Conjugate: $\overline{z} = a - bi$ (flip sign of Im)

Modulus: $|z| = \sqrt{a^2 + b^2}$ (distance to 0)

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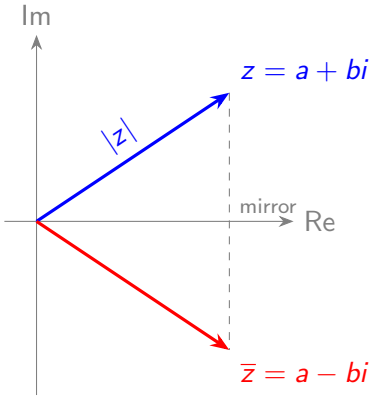
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Modulus: $|z| = \sqrt{a^2 + b^2}$ (distance to 0)

Key identity

$$z\,\overline{z} = (a + bi)(a - bi) = a^2 + b^2 = |z|^2$$

Useful rules: $\overline{z + w} = \overline{z} + \overline{w}$, $\overline{zw} = \overline{z}\,\overline{w}$,
$|zw| = |z||w|$,
$z + \overline{z} = 2\,\mathrm{Re}\,z$, $z - \overline{z} = 2i\,\mathrm{Im}\,z$.



Division: Multiply by the Conjugate

To divide, make the denominator real:

$$\frac{z}{w} = \frac{z\,\overline{w}}{w\,\overline{w}} = \frac{z\,\overline{w}}{|w|^2} \qquad (w \neq 0).$$

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$$\frac{3+2i}{1-4i} = \frac{(3+2i)(1+4i)}{(1-4i)(1+4i)} = \frac{3+12i+2i+8i^2}{1+16} = \frac{-5+14i}{17}.$$

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Handy special case

$$\frac{1}{i} = \frac{\overline{i}}{|i|^2} = -i. \qquad \text{In general } \frac{1}{z} = \frac{\overline{z}}{|z|^2}.$$

Problems

Compute:

1. $(2 - i)(1 + 3i)$
2. $|3 - 4i|$
3. $\dfrac{1}{1 + i}$
4. $\dfrac{}{i(2 + i)}$

Two minutes. Write your answer in the form $a + bi$.

work here

Solutions

1. $(2-i)(1+3i) = 2+6i-i-3i^2 = 5+5i$
2. $|3-4i| = \sqrt{9+16} = 5$
3. $\dfrac{1}{1+i} = \dfrac{1-i}{(1+i)(1-i)} = \dfrac{1-i}{2} = \frac{1}{2} - \frac{1}{2}i$
4. $i(2+i) = 2i+i^2 = -1+2i$, so its conjugate is $-1-2i$

Check (4) another way: $\overline{i(2+i)} = \bar{i} \cdot \overline{2+i} = (-i)(2-i) = -2i+i^2 = -1-2i$.

Polar Form

Describe z by its **length** and **angle**:

$$r = |z|, \qquad \theta = \arg z.$$

Then

$$a = r\cos\theta, \quad b = r\sin\theta, \quad z = r(\cos\theta + i\sin\theta).$$

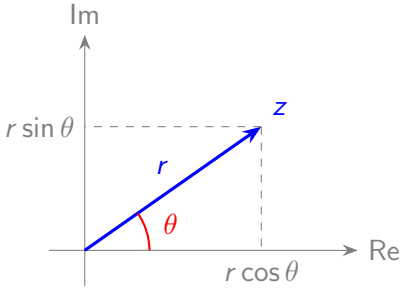
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Example

$1 + i$: $r = \sqrt{2}$, $\theta = \pi/4$.

The angle is only defined up to multiples of 2π. Watch the quadrant: $\theta = \tan^{-1}(b/a)$ alone is wrong when $a < 0$.

Euler's Formula

Euler's formula

$$e^{i\theta} = \cos\theta + i\sin\theta$$

So every complex number can be written

$$z = r\, e^{i\theta}.$$

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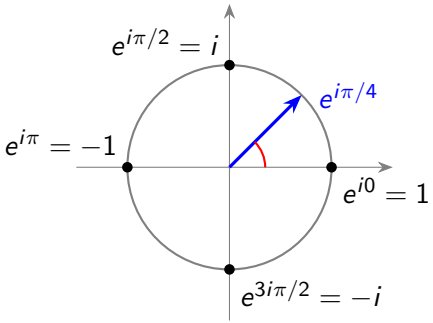
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$|e^{i\theta}| = \sqrt{\cos^2\theta + \sin^2\theta} = 1.$

Famous special case: $e^{i\pi} = -1$.



[Optional] Why Is Euler's Formula True?

Plug $x = i\theta$ into the power series $e^x = \sum_{k \geq 0} x^k/k!$ and sort by powers of i:

$$e^{i\theta} = \underbrace{\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right)}_{\cos\theta} + i\,\underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right)}_{\sin\theta}.$$

The period-4 cycle $i, -1, -i, 1$ is exactly what splits the series into cosine and sine.

For non-math audiences it is fine to take Euler's formula as a definition of $e^{i\theta}$. What matters is how it behaves under multiplication (next slide).

Multiplication = Scale and Rotate

$$r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = (r_1 r_2)\, e^{i(\theta_1+\theta_2)}$$

- Lengths **multiply**.
- Angles **add**.

Multiplication = Scale and Rotate

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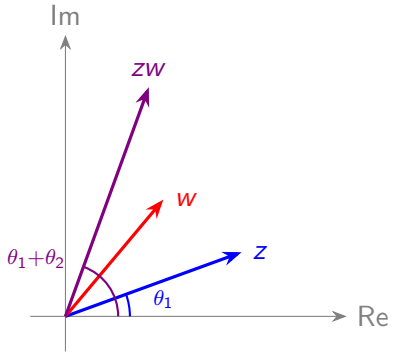
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Much easier than expanding brackets:
$(1+i)^8 = (\sqrt{2})^8 e^{i\,8\pi/4} = 16\, e^{2\pi i} = 16.$



Phases: Why QC Cares

A complex number of modulus 1, $e^{i\varphi}$, is called a **phase**.

- Multiplying by a phase never changes a modulus: $|e^{i\varphi}z| = |z|$.

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$$\left|1+1\right| = 2, \qquad \left|1+e^{i\pi}\right| = |1-1| = 0.$$

Same sizes, different phases $\Rightarrow$ constructive vs destructive **interference**.

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Preview

Amplitudes of a qubit are complex numbers. An overall phase $e^{i\varphi}\psi$ is unobservable (global phase); a phase *between* amplitudes (relative phase) is physical.

Details in the QC Foundations lecture. Today, just note that "length" and "angle" play separate roles.

Roots of Unity

The solutions of $z^n = 1$ are

$$\omega^k, \quad k = 0, 1, \ldots, n-1, \qquad \omega = e^{2\pi i/n}.$$

They sit evenly spaced on the unit circle.

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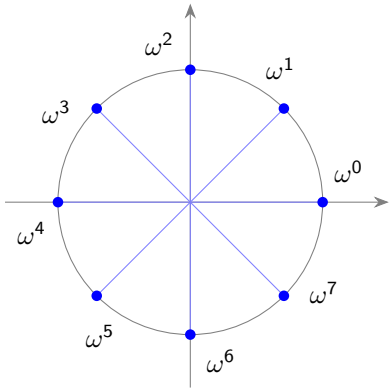
$$\omega^k, \quad k = 0, 1, \ldots, n-1, \qquad \omega = e^{2\pi i/n}.$$

They sit evenly spaced on the unit circle.

They sum to zero (for $n \geq 2$)

$$\sum_{k=0}^{n-1} \omega^k = \frac{\omega^n - 1}{\omega - 1} = 0.$$

This cancellation is the heart of the Quantum Fourier Transform later in the course.



$n = 8$

Problems

1. Write $-1 - i$ in polar form.
2. Compute $(1 - i)^4$ using polar form.
3. Find all z with $z^4 = 1$.
4. Show $|e^{i\alpha} + e^{i\beta}|^2 = 2 + 2\cos(\alpha - \beta)$.

(4) is interference in one line: try it with $z\,\overline{z} = |z|^2$.

work here

Solutions

1. $r = \sqrt{2}$; the point is in the third quadrant, so $\theta = 5\pi/4$ (or $-3\pi/4$): $-1 - i = \sqrt{2}\, e^{5i\pi/4}$.
2. $1 - i = \sqrt{2}\, e^{-i\pi/4}$, so $(1-i)^4 = 4\, e^{-i\pi} = -4$.
3. $\omega = e^{2\pi i/4} = i$: the roots are $1,\ i,\ -1,\ -i$.
4. $|e^{i\alpha} + e^{i\beta}|^2 = (e^{i\alpha} + e^{i\beta})(e^{-i\alpha} + e^{-i\beta}) = 2 + e^{i(\alpha-\beta)} + e^{-i(\alpha-\beta)} = 2 + 2\cos(\alpha - \beta)$.

In (4): equal phases give 4 (constructive), opposite phases give 0 (destructive).

Summary: Complex Numbers

- $z = a + bi = re^{i\theta}$ (Cartesian / polar)
- $\overline{z} = a - bi, \quad z\overline{z} = |z|^2$
- $\frac{1}{z} = \frac{\overline{z}}{|z|^2}$
- Multiply $\Rightarrow$ lengths multiply, angles add
- Phases $e^{i\varphi}$: modulus 1; they cause interference in sums

Take away

A complex number is a length together with an angle. Quantum amplitudes use both.

From Arrows to Lists of Numbers

In $\mathbb{R}^2$ a vector is an arrow, recorded as a list:
$\mathbf{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

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In $\mathbb{C}^n$ a vector is a column of n **complex** numbers:

$$\mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}, \qquad \text{e.g. } \begin{pmatrix} 1+i \\ 2 \end{pmatrix} \in \mathbb{C}^2.$$

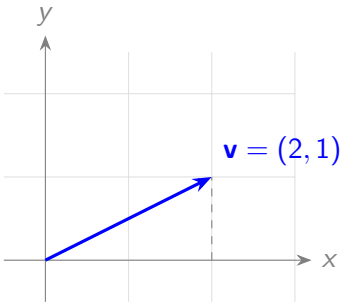
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We cannot draw $\mathbb{C}^2$: it needs 4 real dimensions. So pictures today are $\mathbb{R}^2$ cartoons; the algebra is what carries over.



Operations on $\mathbb{C}^n$

Two operations, both done entry by entry:

Addition

$$\begin{pmatrix} 1+i \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ -i \end{pmatrix} = \begin{pmatrix} 4+i \\ 2-i \end{pmatrix}$$

Scalar multiplication ($\alpha \in \mathbb{C}$)

$$i \cdot \begin{pmatrix} 1+i \\ 2 \end{pmatrix} = \begin{pmatrix} i+i^2 \\ 2i \end{pmatrix} = \begin{pmatrix} -1+i \\ 2i \end{pmatrix}$$

The zero vector is $\mathbf{0} = (0, \ldots, 0)^T$; the negative of $\mathbf{v}$ is $-\mathbf{v} = (-1)\,\mathbf{v}$.

Scalars are now complex. That single change is what makes quantum linear algebra different from what many of you saw in school.

Vector Space over $\mathbb{C}$: Definition

Definition

A set V with addition $\mathbf{u}+\mathbf{v}$ and scalar multiplication $\alpha\mathbf{v}$ ($\alpha \in \mathbb{C}$) is a **complex vector space** if both results stay in V and, for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$, $\alpha, \beta \in \mathbb{C}$:

1. $\mathbf{u}+\mathbf{v} = \mathbf{v}+\mathbf{u}$
2. $(\mathbf{u}+\mathbf{v})+\mathbf{w} = \mathbf{u}+(\mathbf{v}+\mathbf{w})$
3. there is $\mathbf{0}$ with $\mathbf{v}+\mathbf{0} = \mathbf{v}$
4. each $\mathbf{v}$ has $-\mathbf{v}$ with $\mathbf{v}+(-\mathbf{v}) = \mathbf{0}$
5. $\alpha(\mathbf{u}+\mathbf{v}) = \alpha\mathbf{u}+\alpha\mathbf{v}$
6. $(\alpha+\beta)\mathbf{v} = \alpha\mathbf{v}+\beta\mathbf{v}$
7. $\alpha(\beta\mathbf{v}) = (\alpha\beta)\mathbf{v}$
8. $1\,\mathbf{v} = \mathbf{v}$

In words: you can add and scale, and the ordinary rules of algebra hold. That is all.

Examples and Non-Examples

Set (with the usual $+$ and scaling by $\mathbb{C}$)	Vector space?
$\mathbb{C}^n$	✓
All $m \times n$ complex matrices	✓
Polynomials of degree $\leq d$ with complex coefficients	✓
State space of n qubits, $\mathbb{C}^{2^n}$	✓
$\{(z_1, z_2)^T : z_1 = 1\}$ (no zero vector)	✗
$\mathbb{R}^2$ with complex scalars ($i \cdot (1,0)^T \notin \mathbb{R}^2$)	✗
Unit vectors in $\mathbb{C}^2$ ($\mathbf{e}_1 + \mathbf{e}_1$ has length 2)	✗

The last row matters: quantum states are unit vectors, so they do not form a vector space themselves. They live inside one, and we re-normalise when needed.

Linear Combinations and Span

A **linear combination** of $\mathbf{v}_1, \ldots, \mathbf{v}_k$:

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \cdots + \alpha_k \mathbf{v}_k, \qquad \alpha_j \in \mathbb{C}.$$

Linear Combinations and Span

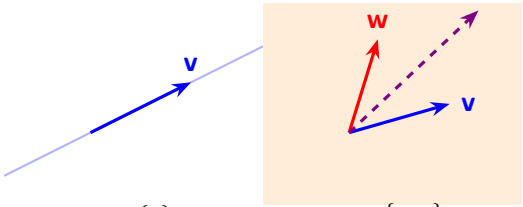
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The **span** is the set of all of them:

$$\text{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}.$$

A quantum "superposition" is exactly a linear combination.



$\text{span}\{\mathbf{v}\}$ = a line

$\text{span}\{\mathbf{v}, \mathbf{w}\}$ = the plane

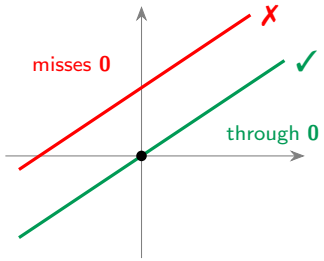
Subspaces

A **subspace** $W \subseteq V$ is a subset that is itself a vector space:

$$\mathbf{u}, \mathbf{w} \in W,\ \alpha \in \mathbb{C} \ \Rightarrow\ \mathbf{u} + \mathbf{w} \in W,\ \alpha\mathbf{u} \in W.$$

- Every span is a subspace.
- Every subspace contains $\mathbf{0}$.

Test: a line through the origin passes; a shifted line fails.



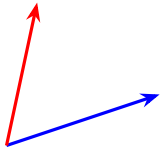
Linear Independence

Definition

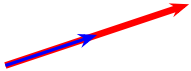
$\mathbf{v}_1, \ldots, \mathbf{v}_k$ are **linearly independent** if

$$\alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k = \mathbf{0} \;\Rightarrow\; \alpha_1 = \cdots = \alpha_k = 0.$$

Otherwise one of them is a combination of the others: it adds nothing new to the span.



independent



dependent
(parallel)

A Complex Twist

Are $\mathbf{u} = \begin{pmatrix} 1 \\ i \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} i \\ -1 \end{pmatrix}$ linearly independent?

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Are $\mathbf{u} = \begin{pmatrix} 1 \\ i \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} i \\ -1 \end{pmatrix}$ linearly independent?

- Over $\mathbb{R}$ (real scalars only): **yes**. No real t gives $\mathbf{w} = t\,\mathbf{u}$.

A Complex Twist

Are $\mathbf{u} = \begin{pmatrix} 1 \\ i \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} i \\ -1 \end{pmatrix}$ linearly independent?

- Over $\mathbb{R}$ (real scalars only): **yes**. No real t gives $\mathbf{w} = t\,\mathbf{u}$.
- Over $\mathbb{C}$: **no**!

$$i \cdot \mathbf{u} = \begin{pmatrix} i \\ i^2 \end{pmatrix} = \begin{pmatrix} i \\ -1 \end{pmatrix} = \mathbf{w}.$$

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Moral

"Parallel" in $\mathbb{C}^n$ means "a *complex* multiple". Vectors that look different can be the same direction.

This is the same phenomenon as global phase: ψ and $e^{i\varphi}\psi$ point in the same complex direction.

Basis and Dimension

Definition

A **basis** of V is a linearly independent set that spans V.

- Every $\mathbf{v} \in V$ is then a *unique* combination $\mathbf{v} = \sum_j c_j \mathbf{b}_j$; the c_j are its **coordinates**.
- All bases have the same size: the **dimension** $\dim V$.

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Standard basis of $\mathbb{C}^n$

$\mathbf{e}_1 = (1, 0, \ldots, 0)^T, \ \ldots, \ \mathbf{e}_n = (0, \ldots, 0, 1)^T, \ \text{so } \dim_{\mathbb{C}} \mathbb{C}^n = n.$

Over $\mathbb{R}$, $\mathbb{C}^n$ would have dimension $2n$ (real and imaginary parts separately). Always ask "dimension over which scalars?"

The Standard Basis of $\mathbb{C}^2$ and Qubits

In $\mathbb{C}^2$:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

and every vector is

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \alpha\,\mathbf{e}_1 + \beta\,\mathbf{e}_2.$$

A qubit's state is such a vector: $\mathbf{e}_1$ and $\mathbf{e}_2$ stand for the classical bit values 0 and 1.

n qubits: dim $= 2^n$

qubits n	dim $\mathbb{C}^{2^n}$
1	2
2	4 (strings 00, 01, 10, 11)
10	1,024
50	$\approx 10^{15}$

A Second Basis: $\mathbf{h}_+$ and $\mathbf{h}_-$

$$\mathbf{h}_+ = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \qquad \mathbf{h}_- = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$$

also form a basis of $\mathbb{C}^2$.

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Same vector, different coordinates:

$$\mathbf{e}_1 = \frac{1}{\sqrt{2}}(\mathbf{h}_+ + \mathbf{h}_-), \quad \mathbf{e}_2 = \frac{1}{\sqrt{2}}(\mathbf{h}_+ - \mathbf{h}_-).$$

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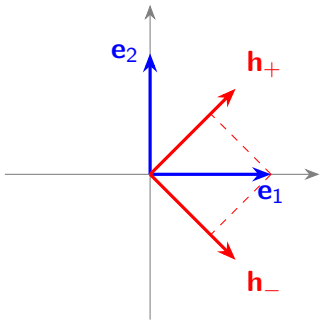
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	in $\{\mathbf{e}_1, \mathbf{e}_2\}$	in $\{\mathbf{h}_+, \mathbf{h}_-\}$
$\mathbf{e}_1$	$(1, 0)$	$(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$



Real cartoon: the entries here happen to be real.

Problems

1. Is $\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$ a basis of $\mathbb{C}^2$?
2. Are $\begin{pmatrix} 1 \\ i \end{pmatrix}$ and $\begin{pmatrix} -i \\ 1 \end{pmatrix}$ independent over $\mathbb{C}$?
3. Write $\begin{pmatrix} 2 \\ 4i \end{pmatrix}$ using $\mathbf{e}_1, \mathbf{e}_2$.
4. Write $\mathbf{e}_2$ in the basis $\{\mathbf{h}_+, \mathbf{h}_-\}$ *without* peeking.

work here

Solutions

1. Yes. $a(1,1)^T + b(1,-1)^T = \mathbf{0}$ gives $a+b=0$ and $a-b=0$, so $a=b=0$; two independent vectors in a 2-dim space form a basis.
2. No: $(-i)\cdot(1,i)^T = (-i,-i^2)^T = (-i,1)^T$.
3. $2\,\mathbf{e}_1 + 4i\,\mathbf{e}_2$.
4. Solve $\mathbf{e}_2 = a\,\mathbf{h}_+ + b\,\mathbf{h}_-$: $\frac{1}{\sqrt{2}}(a+b)=0$, $\frac{1}{\sqrt{2}}(a-b)=1$, so $a=\frac{1}{\sqrt{2}}$, $b=-\frac{1}{\sqrt{2}}$: $\mathbf{e}_2 = \frac{1}{\sqrt{2}}(\mathbf{h}_+ - \mathbf{h}_-)$.

Solving a linear system for (4) works, but is slow. After inner products we will read coordinates off in one step.

Summary: Vector Spaces over $\mathbb{C}$

- $\mathbb{C}^n$: columns of complex numbers; add entrywise, scale by complex α.
- Span = all linear combinations = all *superpositions*.
- Independence and dimension depend on the scalars ($\mathbb{C}$ vs $\mathbb{R}$).
- Basis $\Rightarrow$ unique coordinates; different bases $\Rightarrow$ different coordinates, same vector.
- Qubits: $\mathbf{e}_1, \mathbf{e}_2$ are the standard basis of $\mathbb{C}^2$; n qubits live in $\mathbb{C}^{2^n}$.

Missing so far

We can add and scale, but we cannot yet measure **lengths** or **angles**. Next: inner products.

Warm-up: The Real Dot Product

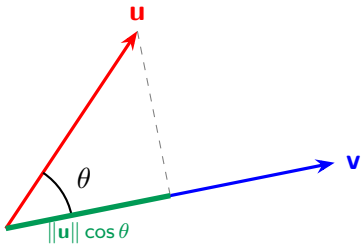
In $\mathbb{R}^n$:

$$\mathbf{u} \cdot \mathbf{v} = \sum_j u_j v_j = \|\mathbf{u}\| \ \|\mathbf{v}\| \cos\theta.$$

One operation gives us

- length: $\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$
- angle: $\cos\theta = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \ \|\mathbf{v}\|}$
- perpendicularity: $\mathbf{u} \cdot \mathbf{v} = 0$

Goal: the same toolkit for $\mathbb{C}^n$. In QC it will also give probabilities.



The Conjugate Transpose $\mathbf{v}^\dagger$

For a column vector $\mathbf{v}$, its **conjugate transpose** $\mathbf{v}^\dagger$ is the row of conjugated entries:

$$\mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \quad \longmapsto \quad \mathbf{v}^\dagger = \overline{\mathbf{v}}^{\,T} = \begin{pmatrix} \overline{v_1} & \cdots & \overline{v_n} \end{pmatrix}.$$

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Also: $(\alpha \mathbf{v})^\dagger = \overline{\alpha}\, \mathbf{v}^\dagger$. The scalar gets conjugated too.

Mnemonic: "dagger" = flip and conjugate. Maths books often write $\mathbf{v}^*$ or $\mathbf{v}^H$ for the same thing.

Why Do We Need the Conjugate?

Try the naive formula $\sum_j v_j v_j$ on $\mathbf{v} = \begin{pmatrix} 1 \\ i \end{pmatrix}$:

$$1 \cdot 1 + i \cdot i = 1 + (-1) = 0.$$

A non-zero vector of "length" zero. Disaster!

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Key point

$\overline{z}\, z = |z|^2 \geq 0$ is what makes lengths real and positive. This is why the complex inner product carries conjugates.

The Inner Product on $\mathbb{C}^n$

Definition

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\dagger \mathbf{v} = \sum_{j=1}^{n} \overline{u_j}\, v_j$$

Row times column: a $1 \times n$ matrix times an $n \times 1$ matrix is a single number.

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Basis vectors of $\mathbb{C}^2$

$\langle \mathbf{e}_1, \mathbf{e}_1 \rangle = \langle \mathbf{e}_2, \mathbf{e}_2 \rangle = 1, \quad \langle \mathbf{e}_1, \mathbf{e}_2 \rangle = 0, \quad \langle \mathbf{h}_+, \mathbf{e}_1 \rangle = \frac{1}{\sqrt{2}}.$

Properties of the Inner Product

For all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and $\alpha, \beta \in \mathbb{C}$:

1. **Conjugate symmetry:** $\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle}$
2. **Linear in the second slot:** $\langle \mathbf{u}, \alpha\mathbf{v} + \beta\mathbf{w} \rangle = \alpha \langle \mathbf{u}, \mathbf{v} \rangle + \beta \langle \mathbf{u}, \mathbf{w} \rangle$
3. **Positive definite:** $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, with equality only for $\mathbf{v} = \mathbf{0}$

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Any function satisfying 1–3 on a vector space is called an inner product; $\mathbb{C}^n$ with $\sum \overline{u_j} v_j$ is the one we use.

Norm (Length)

Definition

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} = \sqrt{|v_1|^2 + \cdots + |v_n|^2}$$

- $\|\mathbf{v}\| \geq 0$, and $\|\mathbf{v}\| = 0 \iff \mathbf{v} = \mathbf{0}$
- $\|\alpha \mathbf{v}\| = |\alpha| \ \|\mathbf{v}\|$
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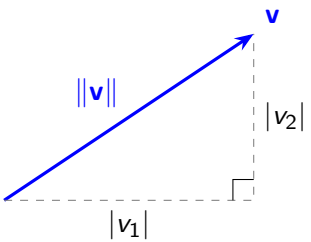
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Example

$$\left\| \begin{pmatrix} 1 \\ 1+i \end{pmatrix} \right\| = \sqrt{1 + |1+i|^2} = \sqrt{1+2} = \sqrt{3}.$$



Pythagoras, with $|v_j|$ in place of v_j.

Normalisation and Quantum States

Any non-zero $\mathbf{v}$ can be scaled to length 1:
$\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$.

Example

$$\mathbf{v} = \begin{pmatrix} 1 \\ 1+i \end{pmatrix} \Rightarrow \hat{\mathbf{v}} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1+i \end{pmatrix}.$$

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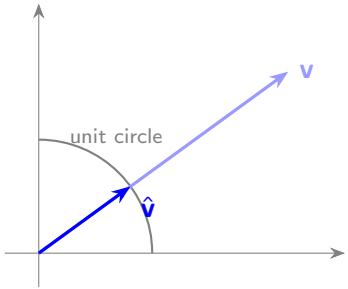
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Quantum states are unit vectors

$\psi = \alpha\,\mathbf{e}_1 + \beta\,\mathbf{e}_2$ with $\|\psi\|^2 = |\alpha|^2 + |\beta|^2 = 1$.

This is exactly the normalisation condition you will see in the QC Foundations lecture.



Two Inequalities You Will Use Constantly

Cauchy–Schwarz

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \, \|\mathbf{v}\|$$

with equality iff $\mathbf{u}, \mathbf{v}$ are parallel ($\mathbf{v} = \alpha \mathbf{u}$).

Triangle inequality

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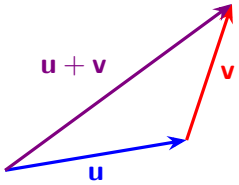
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Why QC cares: for unit vectors, Cauchy–Schwarz gives $0 \leq |\langle \phi, \psi \rangle|^2 \leq 1$, so this quantity can be a probability.



The direct path is never longer than the detour.

Orthogonality

Definition

$\mathbf{u} \perp \mathbf{v}$ iff $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Examples

$\langle \mathbf{e}_1, \mathbf{e}_2 \rangle = 0, \qquad \langle \mathbf{h}_+, \mathbf{h}_- \rangle = \frac{1}{2}(1 \cdot 1 + 1 \cdot (-1)) = 0,$

$\begin{pmatrix} 1 \\ i \end{pmatrix} \perp \begin{pmatrix} i \\ 1 \end{pmatrix}$ (from earlier).

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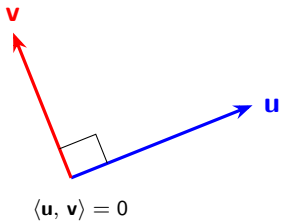
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Pythagoras

If $\mathbf{u} \perp \mathbf{v}$ then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Physically: orthogonal states are perfectly distinguishable. More on this later.



Problems

Let $\mathbf{u} = \begin{pmatrix} 1 \\ i \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 2 \\ 1 - i \end{pmatrix}$.

1. Compute $\langle \mathbf{u}, \mathbf{v} \rangle$ and $\langle \mathbf{v}, \mathbf{u} \rangle$.
2. Compute $\|\mathbf{u}\|$ and $\|\mathbf{v}\|$.
3. Normalise $\mathbf{u}$ and $\mathbf{v}$.
4. Check Cauchy–Schwarz.

Remember: conjugate the first vector.

work here

Solutions

1. $\langle \mathbf{u}, \mathbf{v} \rangle = \overline{1} \cdot 2 + \overline{i}\,(1 - i) = 2 - i(1 - i) = 2 - i + i^2 = 1 - i.$
   $\langle \mathbf{v}, \mathbf{u} \rangle = \overline{\langle \mathbf{u}, \mathbf{v} \rangle} = 1 + i$ (conjugate symmetry).
2. $\|\mathbf{u}\| = \sqrt{1 + 1} = \sqrt{2}, \quad \|\mathbf{v}\| = \sqrt{4 + |1 - i|^2} = \sqrt{6}.$
3. $\hat{\mathbf{u}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad \hat{\mathbf{v}} = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 - i \end{pmatrix}.$
4. $|\langle \mathbf{u}, \mathbf{v} \rangle| = |1 - i| = \sqrt{2} \ \leq \ \sqrt{2} \cdot \sqrt{6} = \sqrt{12}.$ ✓

Orthonormal Bases

Definition

A basis $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is **orthonormal** (ONB) if

$$\langle \mathbf{e}_i, \mathbf{e}_j \rangle = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

i.e. unit length and pairwise orthogonal.

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i.e. unit length and pairwise orthogonal.

Three standard ONBs of $\mathbb{C}^2$

Standard	$\mathbf{e}_1 = (1,0)^T, \quad \mathbf{e}_2 = (0,1)^T$	
Hadamard	$\mathbf{h}_\pm = \frac{1}{\sqrt{2}}(1, \pm 1)^T$	
Circular	$\mathbf{c}_\pm = \frac{1}{\sqrt{2}}(1, \pm i)^T$	*needs complex numbers!*

Coordinates in an ONB: Just Take Inner Products

Expansion formula

If $\{\mathbf{e}_i\}$ is an ONB then for every $\mathbf{v}$

$$\mathbf{v} = \sum_i \langle \mathbf{e}_i, \mathbf{v} \rangle \ \mathbf{e}_i.$$

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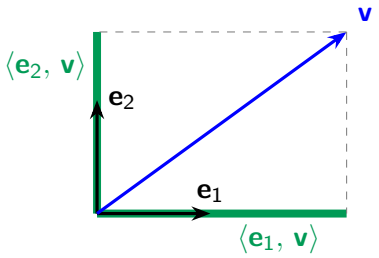
If $\{\mathbf{e}_i\}$ is an ONB then for every $\mathbf{v}$

$$\mathbf{v} = \sum_i \langle \mathbf{e}_i, \mathbf{v} \rangle \, \mathbf{e}_i .$$

Why: write $\mathbf{v} = \sum_i c_i \mathbf{e}_i$ and take the inner product with $\mathbf{e}_j$:

$$\langle \mathbf{e}_j, \mathbf{v} \rangle = \sum_i c_i \langle \mathbf{e}_j, \mathbf{e}_i \rangle = \sum_i c_i \, \delta_{ji} = c_j .$$

No linear system to solve: each coordinate is one inner product.



Worked Example: Rewriting a Vector in the Hadamard Basis

Let $\mathbf{v} = \frac{3}{5}\,\mathbf{e}_1 + \frac{4i}{5}\,\mathbf{e}_2 = \begin{pmatrix} 3/5 \\ 4i/5 \end{pmatrix}$.

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$$\langle \mathbf{h}_+, \mathbf{v} \rangle = \frac{1}{\sqrt{2}}\left(1 \cdot \frac{3}{5} + 1 \cdot \frac{4i}{5}\right) = \frac{1}{\sqrt{2}}\left(\frac{3}{5} + \frac{4i}{5}\right),$$

$$\langle \mathbf{h}_-, \mathbf{v} \rangle = \frac{1}{\sqrt{2}}\left(1 \cdot \frac{3}{5} - 1 \cdot \frac{4i}{5}\right) = \frac{1}{\sqrt{2}}\left(\frac{3}{5} - \frac{4i}{5}\right).$$

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So

$$\mathbf{v} = \frac{3+4i}{5\sqrt{2}}\,\mathbf{h}_+ + \frac{3-4i}{5\sqrt{2}}\,\mathbf{h}_-.$$